

Polygon Interior and Exterior Angle Sums

Interior angle sums, missing interior angles, exterior angles, and finding the number of sides

- Write the equation you set up before you solve it.
 - Every answer on this sheet is a whole number — no rounding is needed.
 - “Regular” means all sides and all angles are equal.
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1. Find the sum of the measures of the interior angles of a hexagon (a polygon with 6 sides).

Answer: _____

2. Find the sum of the measures of the interior angles of a polygon with 11 sides.

Answer: _____

3. A regular polygon has 9 sides. Find the measure of each exterior angle.

Answer: _____

4. Three of the angles of a quadrilateral measure 112° , 87° and 63° . Find the measure of the fourth angle.

Answer: _____

5. Each exterior angle of a regular polygon measures 24° . How many sides does the polygon have?

Answer: _____

6. Four of the angles of a pentagon measure 102° , 118° , 96° and 131° . Find the measure of the fifth angle.

Answer: _____

7. Each interior angle of a regular polygon measures 156° .

(a) Find the measure of each exterior angle. _____

(b) Find the number of sides. _____

8. The interior angles of a polygon add up to 1800° . How many sides does the polygon have?

Answer: _____

9. Marcus is finding the sum of the interior angles of a pentagon. He writes

$$\text{sum} = 5 \cdot 180^\circ = 900^\circ.$$

(a) Find the correct sum of the interior angles of a pentagon. _____

(b) Explain what Marcus did wrong, and say what he should have used instead.

10. In a regular polygon each interior angle is 5 times as large as each exterior angle.

(a) Find the measure of each exterior angle. _____

(b) Find the number of sides of the polygon. _____